Tarkastele tuloksia

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Survey objectives, target group and instructions

Objectives of this survey:
a. To begin to understand the expected added value of National Cancer Data Nodes (NCDNs)
b. To get a first reflection on the functional needs of NCDN (Section 2)
c. To understand the operational needs for Member States (Section 3)

Target group:
This survey is intended for representatives of Member States who are familiar with potential plans and stakeholders related to NCDNs in their respective countries.

Instructions:
Please respond to the survey to the best of your knowledge. You are welcome to consult relevant experts to inform your answers. However, there should be only one response per country.

The survey must be completed in a single session. A PDF copy of the survey is attached to this email for your reference. You may share it with colleagues you consult during the process.

You can set the interface language to English using the language selector in the top right corner.

Initial framework for National Cancer Data Nodes

Currently cancer research and care require a huge amount of complex and multimodal data often not readily available to a researcher or health care professional. Cancer data includes all cancer-related data (omics, imaging, clinical, registry data etc), generated from both health care and research, interpreted along diagnosis, treatment and outcome trajectories.

The aim for National Cancer Data Nodes (eCancer WG) is to facilitate the access of cancer digital assets from multiple sources by offering information and tools to researchers and national cancer data holders that complement EHDS infrastructure capabilities at national level and that leverage on the services already offered by current EU Research Infrastructures.

Furthermore, National Cancer Data Nodes aim to foster collaboration, capacity building and knowledge sharing in the field of cancer data among European cancer research institutions, national cancer research networks, data holders and data nodes, with the goal of accelerating research and innovation and improving patient outcomes.

Section 1: General Information

1. Respondent's name *

Fotis Psomopoulos

2. Respondent's email *

fpsom@certh.gr

3. Country of the respondent *

Greece

4. In what role are you representing your country? *

Representative of the GSRI-mandated NCDN

5. Organization of the respondent *

Centre for Research and Technology Hellas

6. Role in the organization *

Senior Researcher

7. Which stakeholder groups views are you familiar with? *

- [x] Patients
- [x] Clinicians
- [x] Researchers
- [x] Research infrastructures
- [x] Policy makers
- [x] hospital IT staff

8. What is your role/mandate in your country's plans for setting up a National Cancer Data Node (NCDN)? *

CERTH has been mandated by the General Secretariat for Research and Innovation (ref.no. 69040/09.09.2024) as the greek NCDN.

9. Who else should be included as key players of NCDNs from your country in future developments?

IDIKA, EOPYY, Cancer Hospitals (especially the Comprehensive Cancer Centres)

Section 2. National Cancer Data Node Functions

The eCANCER Thematic Working Group identified the following five core functions of an NCDN:

1. Establish **national cancer data nodes** for cancer data holders
2. Improve **data quality** of cancer data at source
3. Ensure all cancer datasets have a **metadata record** in the EU dataset catalogue
4. Develop **common dataset variables** for cancer datasets
5. Provide **tools for cancer data analysis** to be used in secure processing environments (SPEs)

10. Please rank functions 2–5 by dragging them into an order, starting with the most relevant for your country's NCDN and ending with the least relevant. *

1 Develop **common dataset variables** for cancer datasets

2 Improve **data quality** of cancer data at source

3 Ensure all cancer datasets have a **metadata record** in the EU dataset catalogue

4 Provide **tools for cancer data analysis** to be used in secure processing environments (SPEs)

11. Are there additional functions you consider important for your country?

It is important to establish processes and tools that enable seamless linkage between existing cancer databases and the forthcoming National Cancer Registry. In parallel, developing sustainable funding schemes and supporting stakeholder education related to the EHDS and emerging organizational paradigms, such as federated analysis, will be essential. Furthermore, the country should prioritize structured and secure data interconnection across hospitals, research institutes, and registries by providing standardized procedures for data-access agreements, governance frameworks, pseudonymisation workflows, and mechanisms that ensure AI-readiness.

2.1 Establish **National Cancer Data Nodes** for Cancer Data Holders

12. What types of cancer data exist in your country? *

- [x] Clinical data (data collected for healthcare purposes)
- [x] Research data (data collected for the purpose of scientific research)
- [x] Registry data
- [x] Biobank data
- [x] Imaging data
- [x] Genetic data
- [x] several disease-specific databases (hosted and/or supervised by medical societies, collaborative groups and/or individual researche

13. Which types of organizations or data holders store cancer data in your country *

- [x] Hospitals/Clinics
- [x] Universities/Research Institutes
- [x] Registries
- [x] Biobanks
- [x] Private Sector
- [x] Government and state-owned companies (e.g. IDIKA)

14. What types of data and in which format is stored by hospitals/clinics?

Answer if it applies to your country

Hospitals and clinics store a wide range of data types using heterogeneous formats. Clinical data are commonly maintained in tabular structures or relational SQL databases, while biobank data may reside in either SQL or NoSQL systems. Operational and clinical information—such as imaging, hospitalization records, diagnoses, medications, and laboratory information—is often retained in custom or proprietary formats, with only one institution (Papageorgiou General Hospital) currently implementing the OMOP Common Data Model. Omics and other molecular datasets are also present, with NGS data stored in FASTQ, VCF, or processed text files, and additional molecular data maintained in CSV or Excel formats. Metadata associated with deposited datasets may follow standards used by repositories such as GEO or ArrayExpress.

15. What types of data and in which format is stored by universities/research institutes?

Answer if it applies to your country

Clinical data are stored in tabular format or relational sql databases. Biobank data stored in both sql and nosql databases. All other datatypes maintained by hospital/clinics can also be found in universities/research institutes.

16. What types of data and in which format is stored by registries?

Answer if it applies to your country

Clinical data (data collected for healthcare purposes), Genetic data

17. What types of data and in which format is stored by biobanks?

Answer if it applies to your country

It varies by biobank, and it should be noted that there are various levels of alignment to the biobank required standards across the existing ones (very few fully comply).

18. What types of data and in which format is stored by private sector?

Answer if it applies to your country

Types are already listed under Q12. Generally speaking, hospital data and lab results data are managed by various private companies, and there is no widely used format.

19. What types of data and in which format is stored by other organizations or data holders?

Answer if it applies to your country

Other organizations and data holders maintain a variety of datasets, including clinical data generated through healthcare services, research data produced in scientific studies, and specialized imaging and genetic data. IDIKA is the principal national entity managing healthcare information, including cancer-related datasets, and it typically employs custom data formats for storing and processing these resources.

20. Are there already infrastructures or players in your country that support or connect cancer data holders, such as national nodes of European Research Infrastructures? *

- (●) Yes
- ( ) No
- ( ) I don't know

21. What in infrastructures or players supporting or connecting cancer data holders, such as national nodes of European Research Infrastructures, exist in your country? *

- [x] BBMRI-ERIC
- [x] ELIXIR
- [x] EATRIS-ERIC
- [ ] Euro-BioImaging ERIC
- [x] ECRIN-ERIC
- [x] Federated EGA (FEGA)
- [ ] Muu

22. Are these infrastructures linked to other EU projects in your country?

Please name the projects in your answer.

Although all the RIs listed in Q21 are involved, there is still a gap between the RIs and the practical support required by the key stakeholders. Also, there is a coherent effort to link some of them to the EOSC-related project as well as efforts like EMA DARWIN (through OHDSI).

23. Are there already infrastructures or players in your country that support or connect cancer data holders, such as national nodes of EU-funded data infrastructure initiatives? *

- (●) Yes
- ( ) No
- ( ) I don't know

24. What infrastructures or players exist in your country that support or connect cancer data holders, such as national nodes of EU-funded data infrastructure initiatives? *

- [ ] GDI
- [x] EUCAIM
- [x] EOSC4Cancer
- [ ] CanSERV
- [x] B1MGplus
- [ ] TEHDAS2
- [x] CANDLE
- [ ] UNCAN-CONNECT
- [ ] HealthData@EU Pilot
- [ ] Muu

25. Do the existing cancer data holders in your country collaborate? *

- ( ) Yes
- (•) No
- ( ) I don't know

26. Is there expertise in your country in data security to advise data holders on relevant issues within the EHDS? *

- (•) Yes
- ( ) No
- ( ) I don't know

27. Please describe the expertise there is in your country in data security to advise data holders on relevant issues within the EHDS. *

There are many teams working on cybersecurity and federated analysis (a key concept in the context of EHDS). These teams could certainly act as advisors for data holders.

28. What processes and mechanisms exist in your country to identify, interact with and/or coordinate national data holders? *

There is no currently established and widely applied process. There is a Health Data Access Body (HDAB) currently being set up.

29. Do you consider your country has the ability to participate in a European network of cancer data nodes? *

- (●) Yes
- ( ) No
- ( ) I don't know

2.2 Improve **Data Quality** of Cancer Data at Source

30. Are there any initiatives/projects in your country aiming to improve data quality at source for **clinical data**? *

- (●) Yes
- ( ) No
- ( ) I don't know

31. Please name the initiatives/projects aiming to improve data quality at source for **clinical data** in your country. *

Yes, partially. There are some limited projects currently being funded by the "Recovery and Resilience Fund – Greece 2.0". One of the most relevant ones is the "Digitalization of Health Records" - https://greece20.gov.gr/en/?projects=psifiopoiisi-ton-archeion-toy-kratoys-16778-6

32. Are there any initiatives/projects in your country aiming to improve data quality at source for **research data**? *

- [x] Yes
- [ ] No
- [ ] I don't know

33. Please name the initiatives/projects aiming to improve data quality at source for **research data** in your country. *

Primarily led by the Precision Medicine Networks in Greece. The OHDSI-GR (https://www.ohdsi-europe.org/index.php/national-nodes/greece) as well as the ELIXIR-GR Communities are tackling technical perspectives (clinical data and life science data respectively).

34. Are there any initiatives/projects in your country aiming to improve data quality at source for **registry data**? *

- [ ] Yes
- [x] No
- [ ] I don't know

35. Are there any initiatives/projects in your country aiming to improve data quality at source for **biobank data**? *

- [x] Yes
- [ ] No
- [ ] I don't know

36. Please name the initiatives/projects aiming to improve data quality at source for **biobank data** in your country. *

BBMRI-GR.

37. Are there any initiatives/projects in your country aiming to improve data quality at source for **imaging data**? *

- ( ) Yes
- ( ) No
- (•) I don't know

38. Are there any initiatives/projects in your country aiming to improve data quality at source for **genetic data**? *

- (•) Yes
- ( ) No
- ( ) I don't know

39. Please name the initiatives/projects aiming to improve data quality at source for **genetic data** in your country. *

Precision Medicine Networks (https://oncopmnet.gr/)

40. What functions are required to improve cancer data quality at source in your country? *

There are some databases hosted by various institutions but there isn't a usable registry yet. There is a new registry that has just been setup (February 2025) but it does not (yet) include a lot of data.

41. What support or guidance would you need to ensure high-quality, FAIR-compliant data from the outset? *

The key challenge is primarily lack of funding. The technical expertise already exists in the country.

42. What are the bottlenecks and opportunities? *

Funding is a huge issue. Furthermore, digitalization of hospitals and cancer centres might also not be there.

2.3 Ensure All Cancer Datasets Have a **Metadata Record** in the EU Dataset Catalogue

43. In your country, is there any initiative defining a national metadata record of all cancer data sets? *

- (●) Yes
- ( ) No
- ( ) I don't know

44. Please name the initative(s) defining a national metadata record of all cancer data sets in your county. *

Jane-2, ELIXIR-GR. Mainly effort only at local projects – primarily building on existing standards such as HealthDCAT-AP and MIABIS.

45. What standard is used in the initiative(s) defining a national metadata record. Please select all that apply. *

- [ ] CDISC
- [ ] The 1 +MG Minimal Dataset for Cancer
- [ ] GDI Minimal Cancer Dataset
- [ ] DigiOne MEDOC
- [x] DCAT-AP
- [x] I don't know
- [ ] Muu

46. What is the status of the initiatives?

-

47. What are the bottlenecks and opportunities? *

N/A

2.4.1 Develop **Common Dataset Variables** for Cancer Datasets - **Structural Standardization**

Definition of structural standardization: The organization of data elements and the definition of their relationships within a data model.

48. Are there any national initiatives in your country that define common dataset variables for cancer data on a structural level (e.g. national cancer registry schemas covering patient demographics, tumor characteristics, treatment, outcomes)? *

- ( ) Yes
- (•) No
- ( ) I don't know

49. Is your country part of any of the following or other EU initiatives defining structural data standardization? *

- [x] HL7 Europe / HL7 FHIR
- [ ] GA4GH
- [x] DarwinEU
- [x] EHDEN
- [ ] OpenEHR
- [ ] No
- [x] OHDSI-GR, ELIXIR-GR, BBMRI

50. What is the status of these structural standardization EU initiatives in your country? *

Primarily at local efforts only. For example, the Papageorgiou General Hospital (PGH) with the support of the Institute of Applied Biosciences at the Centre for Research and Technology Hellas (INAB|CERTH) participates in EMA DARWIN. Furthermore, INAB|CERTH actively participates in EHDEN foundation and relevant activities, in addition to other teams actively participating on HL7 Europe.

51. Which institutions and/or organizations are involved within the country in the initiatives?

Papageorgiou General Hospital (PGH), INAB|CERTH, IDIKA

52. Which standards or data models are currently used or under consideration to standardize cancer data on a structural level in your country? *

- [ ] HL7 FHIR
- [ ] OMOP CDM
- [ ] OpenEHR
- [x] None
- [ ] Muu

53. In which areas do you anticipate that existing standards are insufficiently specific for cancer data, and what gaps should be addressed?

While the OMOP Common Data Model is currently being applied to cancer data at Papageorgiou General Hospital in Thessaloniki, significant limitations remain. A substantial portion of cancer-related information has not yet been fully digitalized, constraining the applicability of existing standards. Moreover, within OMOP-CDM itself, several oncology-specific gaps require further development, including more precise approaches for representing metastases at specific anatomical sites and the standardized mapping of genetic variants. Despite these limitations, OMOP-CDM remains the most advanced and promising framework for enabling the secondary use of cancer data, provided that these outstanding issues are systematically addressed.

54. What are the main bottlenecks and opportunities in implementing structural data standards for cancer data? *

Funding is always an issue. Another frequently discussed subject is the legal base on which this data are used, which is not yet clear in Greece.

2.4.1 Develop **Common Dataset Variables** for Cancer Datasets - **Semantic Standardization**

Definition of semantic standardization: The adoption of a common (standardized) nomenclature that enables consistent interpretation of data across systems.

55. Are there national initiatives in your country that define common dataset variables for cancer data at the semantic level? *

- ( ) Yes
- ( ) No
- (•) I don't know

56. Is your country involved in the following or other EU or international initiatives for semantic data standardization? *

- [x] UNCAN.eu
- [x] EUCAIM
- [ ] IDEA4RC
- [x] TEHDAS2
- [ ] None
- [ ] Muu

57. Which semantic standards, ontologies or terminologies are currently used to standardize data on a semantic level in your country? *

- [x] SNOMED CT
- [x] ICD-O
- [x] LOINC
- [ ] None
- [x] DRGs

58. Which semantic standards, ontologies or terminologies would you consider adopting for specific data types (clinical data, imaging data, genomics data, registry data etc)? *

This could be considered as a secondary point to be resolved, as there are already activities (such as OHDSI Athena) which map relevant dictionaries and interlinked them.

59. What are the main bottlenecks and opportunities in implementing semantic data standardizations for cancer data? *

Interoperability activities (such as ELIXIR and the OHDSI Athena) could act as catalysts for addressing potential bottlenecks on this.

2.5 Provide **Tools for Cancer Data Analysis** to be Used in Secure Processing Environments*

*Secure processing environments (SPEs)

60. In your country, is there any initiative defining SPEs for cancer data analysis? *

- (●) Yes
- ( ) No
- ( ) I don't know

61. Please describe the initiatives and their status. *

The INCISIVE project uses federated learning as a form of SPE for cancer data analysis, enabling AI model training on sensitive data without transferring it. INCISIVE's federated learning infrastructure is fully operational, supporting GDPR-compliant decentralized processing of de-identified imaging and clinical data, integrated AI analysis tools, and both central and distributed repositories.

62. Are there defined software tools in the SPEs in cancer research used in your country?

- (●) Yes
- ( ) No
- ( ) I don't know

63. Are there established standards or requirements for the quality of the software tools? *

- (●) Yes
- ( ) No
- ( ) I don't know

64. What standards are used? *

- [x] EOSC Software Quality recommendations (EVERSE RSQKit, Software Quality)
- [x] IEEE Standard 730-2014
- [ ] I don't know
- [ ] Muu

65. Is your country actively participating in EU initiatives preparing for EHDS (including Digital Business Capabilities, like analytics for Secure Processing Environments) such as TEHDAS2, Community of Practice for Secondary Use of Health Data etc? *

- [x] Yes
- [ ] No
- [ ] I don't know

66. Please describe the initiative(s) and their status. *

IDIKA (which participates in TEHDAS2) are in the process of providing relevant recommendations which would act as the basis of software requirements for SPEs and other relevant developments.

67. What standards are used in these initiative(s)? *

OMOP-CDM is the most actively discussed technical standard to be used.

68. Are there local EHDS implementation projects in your country? *

- [x] Yes
- [ ] No
- [ ] I don't know

69. Are specific Secure Processing Environments (SPEs) intended to be used there? *

- [ ] Yes
- [ ] No
- [x] I don't know

70. What are the bottlenecks and opportunities? *

There is a managerial/operational risk regarding how many things are left on the national nodes to be decided without the overall harmonization at the EU level.

71. What kind of infrastructures, tools and platforms would be most useful to support cancer data analysis and sharing in your country? *

Having a registry and/or hospital cancer data mapped in OMOP-CDM and an SPE available could be a very good first step.

72. What mechanisms are in place to ensure secure data access and present re-identification?

Nothing specific yet.

2.6 Possible Additional Function: Collecting Data from **Diverse Sources**

One additional function of an NCDN could be collecting data either physically or virtually on a national level from diverse sources such as cancer registries, hospital information systems, pathology and genomics labs, imaging archives (e.g. PACS) and clinical trials and research cohorts.

73. Is collecting data from diverse sources an important possible additional function for a NCDN in your opinion? *

- (●) Yes
- ( ) No

74. Please explain the reasoning behind your response.

Collecting data from diverse sources would be a highly valuable function for a NCDN. Integrating information such as insurance records, claims data, pharmacy data, and sick-leave datasets would substantially enhance analytical depth and improve understanding of patient pathways and outcomes. Furthermore, establishing even a high-level overview or mapping of available data sources would represent a significant advancement, as it would streamline study setup, reduce effort, and conserve resources.

2.7 Possible Additional Function: Community Engagement

75. In your Member state, is there already an initiative bringing national cancer data related activities together? *

- ( ) Yes
- (●) No
- ( ) I don't know

76. Can you name people or organizations in your country that you consider would be important in developing the NCDN functions at a later stage in your country?

INAB-CERTH, IDIKA, ODIPY, ELLOK, EOPE, EAE, HPHO

Section 3: NCDN Operations

Questions of this section start on next page

3.1 General Stage of NCDN Development

77. Does your country have a national cancer data strategy? *

- ○ Yes
- ◉ No
- ○ I don't know

78. In what stage is the preparation of an NCDN in your country? *

The role of INAB|CERTH as an NDCN has been mandated by GSRI (ref.no. 69040/09.09.2024), but operationalization is still not in place.

79. What are your key considerations for the establishment of an NCDN? *

Alignment to similar structures in other member states, as well as reaching a consensus at the national level.

80. To your knowledge, is your country currently developing or planning to develop an NCDN in alignment with the EHDS development? *

- ◉ Yes
- ○ No
- ○ I don't know

81. How will the national laws (e.g. EHDS regulations) influence NCDN operation?

Ei vastausta.

3.2 Key players in NCDN governance

82. Please list key players in governance and operations of a potential NCDN in your country (e.g. ministries, HDABs, RIs, cancer data holders). *

Ministries of Health and Digital Governance (potentially, also Ministry of Education), GRECRIN (national node of ECRIN), ODIPY, Relevant Medical Societies (Medical Oncology - EOPE, Hematology-EAE, Pediatric Hematology and Oncology-HPHO), relevant research infrastructures (ELIXIR).

83. Are the key players the same or associated with institutions who are responsible for EHDS implementation in your country? *

- (●) Yes
- ( ) No
- ( ) I don't know

84. In case you already know, please describe how will the decisions be made about data access, prioritization, or oversight in your country's NCDN?

Ei vastausta.

85. Who do you think is the most appropriate person or organization to answer the questions in the NCDN Operations section, and are there others you believe should also be involved?
Please list all relevant individuals or organizations.

Ei vastausta.